

**THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS**

MSBA7032 Quantitative Trading / AI in Finance

GENERAL INFORMATION	
Instructor: Prof. Weichen Wang Email: weichenw@hku.hk Office: KK 1319 Phone: 3917 1617 Consultation times: Before/after class or by appointment Tutor: TBC Pre-requisites: MSBA7002 Business Statistics MSBA7027 Machine Learning is not required, but knowledge on machine learning is preferred Co-requisites: None Mutually exclusive: N/A Course Website: Moodle The following programming languages are used in this course: Python	
COURSE DESCRIPTION	
Quantitative trading is an interdisciplinary subject involving a combination of techniques in statistical analysis, feature engineering, time series forecasting, machine learning and portfolio optimization. Through this course, students will get exposure to basic asset pricing for financial markets and how state-of-the-art quantitative techniques may be applied in real trading decisions.	
COURSE OBJECTIVES	
1. Understand stylized features of financial markets and the role of quantitative trading 2. Lay out a unified framework where different quantitative techniques fit into the big picture of quant trading 3. Learn and review some useful statistical tools, machine learning methods and optimization algorithms 4. Obtain knowledge of cross-sectional asset pricing and portfolio allocation 5. Gain hands-on experience in equity market forecasting, back-testing and performance evaluation	
PROGRAMME LEARNING OUTCOMES	
PLO1: Acquisition and internalization of knowledge of the programme discipline PLO2: Application and integration of knowledge PLO3: Inculcating professionalism and leadership PLO4: Developing global outlook PLO5: Mastering communication skills	
COURSE LEARNING OUTCOMES	
Course Learning Outcomes	Aligned Programme Learning Outcomes

CLO1: Clearly identify and formulate the relevant problem CLO2: Select and use effective methods to address the problem CLO3: Use software tools to provide solution to the issue at hand CLO4: Communicate the solution effectively		PLO1 PLO2 PLO2, 4 PLO3, 5	
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures		30	25%
T&L2. Assignments, tutorials		60	50%
T&L3. Self-study, group discussion		30	25%
Total		120	100%
Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Participation	Attendance & discussions	20%	CLO1, 2, 3, 4
A2. Assignments	Effort and accuracy	40%	CLO1, 2, 3, 4
A3. Final project	Effort and accuracy	40%	CLO1, 2, 3, 4
Total		100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments		
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good performance in assignments		
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable performance in assignments		
D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable performance in assignments		

F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments and the Exam	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE
Lecture 1: Introduction to Quantitative Trading • Stylized features of asset returns • Market efficient hypothesis Lecture 2-4: Factor Analysis • CAPM and multi-factor pricing model • Fama-French, Barra and PCA approaches • Review of classical factors • Alpha vs Beta, factor risk premium vs factor timing Lecture 5: Mid-module Presentation • Research plan and basic idea Lecture 6-8: Developing Trading Strategy • Feature engineering: technical analysis, fundamental analysis and alternative data • Return dynamics modelling: ARIMA • Shrinkage and regularization: penalized least squares • Nonlinearity and interactions: machine learning and deep learning Lecture 9: Portfolio Optimization and Risk Assessment • Large covariance matrix estimation • Optimization algorithms, trading constraints and back-testing Lecture 10: Execution, High Frequency Trading and Other Markets
REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)
To be announced on Moodle.
MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE
☒ conducting mid-term survey in addition to SETL around the end of the semester ☐ Online response via Moodle site ☐ Others: _____ (please specify)
COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)
An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.
ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)
Assignments will be submitted in class before the lectures. No late assignment will be accepted. Lecture notes and self-learning materials will be uploaded on Moodle. Good questions and discussions on Moodle will be awarded participation grades.